

“WE ARE BUILDING What Could Be The First **SINGLE-CHIP** Battery Management System”

Rajesh Gupta from MOSart Semiconductor speaks with Nidhi Agarwal and Nikita Kumari from Electronics For You about building a single-chip battery management architecture, continuous battery health monitoring, and the roadmap for India-led semiconductor design.



What is the core innovation in your single-chip battery management system IC?

Our innovation lies in integrating multiple battery management system (BMS) functions into a single IC, replacing the multi-chip approach used today. We are also introducing a new patent-pending feature called inline Tru-EIS (electrochemical impedance spectroscopy). This reduces system complexity, lowers cost, improves accuracy, and shortens development time.

Conventional BMS designs rely on multiple ICs mounted on a PCB, each handling separate sensing, control, and safety functions. The inline Tru-EIS acts as a continuous health monitor for each cell in the battery, monitoring the rate of electrochemical reactions within each cell.

This continuous monitoring enables improved battery life, higher reliability, and early fault detection, potentially making it a game-changer for electric vehicles (EVs) and battery energy storage system (BESS) applications.

In the ML1026 architecture, these functions are consolidated into a single chip. By allowing all functional blocks to share internal memory rather than communicate externally, the design reduces wiring complexity, communication delays, and power consumption while improving speed, reliability, and overall system efficiency.

What are the main functional capabilities integrated inside the BMS chip?

Thanks to

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